

**IDAHO NATIONAL LABORATORY
NATIONAL REACTOR INNOVATION CENTER (NRIC)**



NUCLEAR ENERGY LAUNCH PAD

REQUEST FOR APPLICATION (RFA) NUMBER: INL-NRICRFA01

RFA Issue Date:	April 29, 2026
Initial Submission Deadline for Questions:	Potential applicants must submit RFA questions by June 19, 2026, at 5:00 PM MDT. NRIC will answer all questions received by that date and may answer questions submitted after that date at its discretion. All questions should be directed to: NRIC@inl.gov
Industry Day:	May 19, 2026, virtual attendance only. Refer to the NRIC website for Industry Day details and registration requirements. See link provided Launch Pad Industry Day Information
Initial Submission Deadline for Applications:	Applications must be submitted by July 08, 2026, at 5:00 PM MDT to be considered for the first selection round. After this date, applications may be submitted at any time and will be reviewed periodically for selection as soon as reasonably possible.
Initial Anticipated Selections:	NRIC anticipates announcing initial selections of participants by August 19, 2026, but will be dependent upon the number of applications received. NRIC may announce further selections of participants on a rolling basis after that date.
* After initial applications the RFA will remain open continuously and applications will be reviewed periodically for future selections. Time for future reviews and selections will be determined.	

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A. Request for Application Details

INTRODUCTION

The U.S. Department of Energy (DOE) and the National Reactor Innovation Center (NRIC) are establishing the Nuclear Energy Launch Pad, an NRIC initiative aimed at promoting the rapid development and implementation of private industry deployments of a variety of advanced nuclear technologies. The NRIC Launch Pad initiative builds upon DOE’s Reactor and Fuel Line Pilot Programs' success, creating an ecosystem to expedite advanced nuclear technology deployment using flexible technical and regulatory frameworks on federal and non-federal lands, through two distinct pathways, Launch Pad Idaho National Laboratory (INL) and Launch Pad United States of America (U.S.A).

DOE established the Reactor Pilot Program in June 2025, in response to Executive Order 14301, to leverage DOE authorization with the goal of achieving criticality in three reactors by July 4, 2026. Additionally, in August 2025, DOE established the Fuel Line Pilot Program, to support the Reactor Pilot Program and leverage DOE authorization to establish a domestic nuclear fuel supply chain for testing new reactors.

Given DOE’s Pilot Programs' mission success, DOE plans to extend these opportunities using the NRIC Launch Pad. Launch Pad will build on and enhance the DOE's Pilot Programs by creating a physical space and framework to allow continued nuclear innovation. As with the Pilot Program projects, Launch Pad will unlock private capital and enable fast tracking towards commercial licensing.

Types of nuclear facilities envisioned for the Launch Pad initiative include, but are not limited to, nuclear reactors, fuel fabrication, other fuel cycle facilities (i.e., fuel recycling, enrichment, conversion/deconversion facilities), and other supporting facilities.

PROCESS

Initially, applicants are to submit their applications in accordance with the deadlines on the cover page under this Requests for Application (RFA). This Launch Pad RFA opportunity is expected to open

continuously for new applicants from then on with regular application review and selection periods to be determined. NRIC will maintain flexibility and perform review and selection intermittently as needed and as resources allow. Applications will be evaluated based on the details contained in the Technical Volume found in Section F, Application and Submission Information, and the application review found in Section G, Applicant Review Information. NRIC, in coordination with DOE, will administer reviews and DOE reserves the right to select all, some, or none of the applications received in response to this RFA. Priority for award will focus on applicants who can achieve earlier deployment.

Some applicants who have submitted applications to DOE's reactor or fuel pilot programs will be notified and transitioned into the Launch Pad program and will be afforded the opportunity to update their application as desired for the Launch Pad initiative. If selected to participate in the Launch Pad initiative, applicants will receive notification to initiate the applicable agreements.

An acceptable advanced nuclear facility must meet the following thresholds:

1. *Sufficiently mature design* to support imminent development and submittal of safety basis documentation, including a final documented safety analysis to support operations.
2. *Established nuclear material plans* that include disposition or lifecycle pathways for all nuclear materials utilized or generated during testing, with additional specifics appropriate to the facility:
 - a. For reactor facilities, plans should incorporate qualified fuel forms with identified fuel fabrication plans,
 - b. For reactor, enrichment, conversion/deconversion and fuel fabrication facilities, plans should identify procurement pathways for feedstock material(s) and critical-path equipment/parts,
 - c. For fuel recycling facilities, a framework for obtaining used/spent nuclear fuel and a detailed description of expected product and waste streams.
3. *Adequate financial resources* and mature supply chain to complete design, build, commissioning, operation, and decommissioning, as appropriate
4. *Execution readiness* including for procurement/manufacture of all materials, systems and components, and qualified staffing to support all project phases
5. *Developed commercialization plan* for next steps and plans for long-term commercialization, including plans for commercial licensing, future tests and demonstrations and their locations.

STATUTORY AUTHORITY

Statutory authority for this program includes but is not limited to:

42 U.S.C. § 15801 et seq. – National Energy Policy and Programs

42 U.S.C. § 7151 – General transfers

42 U.S.C. § 7256(g) – Contracts, leases, etc., with public agencies and private organizations and persons, Additional Authorities

42 U.S.C. § 2140(a) – Exclusions from license requirement

42 U.S.C. § 2011 et seq. – Atomic Energy Act of 1954

42 U.S.C. § 5801 et seq. – Energy Reorganization Act of 1974

REGULATORY AUTHORITY

Regulatory authority for this program includes but is not limited to:

- 2 C.F.R. Part 930 – Other Transaction Agreements
- 10 C.F.R. Part 820 – Procedural Rules for DOE Nuclear Activities
- 10 C.F.R. Part 830 – Nuclear Safety Management
- 10 C.F.R. Part 835 – Occupational Radiation Protection
- 10 C.F.R. Part 840 – Extraordinary Nuclear Occurrences
- 10 C.F.R. Part 850 – Chronic Beryllium Disease Prevention Program
- 10 C.F.R. Part 851 – Worker Safety and Health Program

B. Program Description

NRIC is establishing a flexible ecosystem for developers to access infrastructure, expertise, and services essential for the siting, construction, and operation of their nuclear facilities.

The Launch Pad initiative provides industry partners with two distinct pathways:

LAUNCH PAD INL

The Launch Pad INL covers approximately 2,000-acres, which will be divided into several plots for private nuclear technology developers, including those working on advanced reactors, fuel fabrication, recycling, enrichment, and other advanced nuclear technology deployment at INL.

Overall benefits:

- Land is suitable for different nuclear applications and regulatory deployment structure.
- Accelerated identification, allocation, and assessment of sites.
- Proximity to existing utilities and services at INL.
- Direct access to specialized nuclear expertise, as needed.
- Assistance in navigating the complexities of nuclear regulations.
- A flexible contract framework with DOE and/or Battelle Energy Alliance, LLC (BEA), DOE's management and operating (M&O) contractor for INL.

LAUNCH PAD U.S.A.

Launch Pad U.S.A. will facilitate projects leveraging DOE authorization at other DOE sites outside INL as well as non-federal sites. This pathway will build upon the successes of DOE's pilot programs and continue to offer the ability for DOE to authorize the operation of nuclear reactors and fuel cycle facilities outside of the national laboratories.

Overall benefits:

- Remote or project-specific access to specialized nuclear expertise at INL or other national laboratories, as needed.

- Assistance in navigating the complexities of nuclear regulations for DOE-authorization.
- A flexible contract framework with DOE and its contractors, as applicable.
- High flexibility to leverage unique regional or project-specific advantages.

Launch Pad INL and Launch Pad U.S.A. facilities will serve research, development, and demonstration purposes. Although Launch Pad will not support commercial deployment projects which are licensed by the Nuclear Regulatory Commission (NRC), companies desiring to site those commercial projects at the INL site should contact DOE for further discussion about land availability. Launch Pad, however, may include projects which begin under DOE authorization and plan to later transition to NRC licensing or other commercialization opportunities. Delivery of qualified test reactor generated electrical power to non-experimental equipment and/or onto the United States Electrical Grid for purchase or nuclear material produced and provided to commercial stakeholders/customers is outside of this RFA and must seek NRC licensing.

C. Agreement Instrument

Selected Launch Pad INL: Applicants may contract with DOE using an Other Transaction Authority (OTA) agreement. DOE will utilize additional authorities under 42 U.S.C. § 7256(g) to implement OTAs under this RFA.

Applicants interested in Launch Pad INL who are selected shall contract with the INL M&O contractor through a Strategic Partnership Project (SPP) agreement, or other appropriate agreement type, for negotiated scope items that support Launch Pad integration between BEA and the Applicant. This partnership provides access to INL's facilities, expertise, research resources, and development support, or may be limited to specific Launch Pad interface requirements. The type of deployment will further define the most applicable agreement instrument(s) for INL deployment.

Selected Launch Pad U.S.A.: Applicants may require an OTA with DOE and shall contract with the INL M&O contractor through an SPP agreement, or other appropriate agreement type. Other agreements may be necessary for Launch Pad U.S.A. but are the responsibility of the Applicant and are not defined in this RFA.

These agreement instruments provide maximum flexibility and efficiency to implement this program in a manner that will allow speed and agility of deployment. It also enables DOE, pursuant to the Atomic Energy Act (AEA), to authorize subject nuclear facilities under DOE's sufficient control. This includes, for example, reactor facilities "under contract with and for the account of" the Department, in accordance with 42 U.S.C. § 2140. Accordingly, facilities built and operated pursuant to the Launch Pad initiative will not require initial NRC licensing. DOE-approved facility designs can help streamline the transition to future NRC licensing.

The terms of these agreements are crafted to streamline regulatory requirements and limit intellectual property (IP) rights to the Government. NRIC and DOE are not seeking to obtain legal rights to any intellectual property such as title to inventions, copyrighted works, or software arising out of this project aside from particular data arising out of the project as discussed below. It is therefore anticipated that agreement terms are close to final and can be implemented as soon as possible after selection.

D. Project Information

I. FUNDING

Development of the application is the financial responsibility of the Applicant. NRIC and DOE will not be providing funding for projects selected as part of Launch Pad and will not be providing funding or in-kind contributions for cost sharing. If selected, the Applicant will be responsible for the costs associated with, but not limited to, designing, manufacturing, constructing, operating, decommissioning, and waste streams related to its facility(s). All industry deployments at a Launch Pad site will be privately funded by the respective Applicant. DOE direct salary costs to review documents necessary for DOE authorization may be the responsibility of the Applicant. DOE/NRIC/INL costs associated with DOE authorization or other activities will be borne by the Applicant. Contributory funds may be required of the Applicant pursuant to the terms and conditions of its respective agreement(s). Contributory funds will be required for federal and contractor travel to support on-site assessments and will be required for contracted personnel to support DOE activities.

II. LOCATION

The Launch Pad INL is an approximately 2,000-acre site located near the INL's Central Facilities Area (CFA) that will be parceled into multiple plots. Some alternative INL lands are identified for additional use if required following review of the applicants INL deployment needs. Final land allocation assignments will be made by NRIC, in coordination with DOE and BEA land use planning, with the intent to optimize use of the space. Launch Pad U.S.A. Applicants will be responsible for proposing and procuring a suitable site in the United States for construction, testing, operation, and decommissioning of their facility(s) to the extent possible under existing United States law and waste frameworks. This includes all site characterization activities to support final site selection. (see Section IX. 1. NEPA)

III. NUCLEAR MATERIAL PROCUREMENT

The Applicant is responsible for procuring all nuclear material feedstock for all facility types.

1. Enrichment facility Applicants should source feedstock domestically and utilize domestic processing technologies to be considered for the Launch Pad initiative. Enriched uranium products must be restricted to peaceful, civilian uses.
2. Fuel fabrication facility Applicants are responsible for preparing a plan which will support the recycling of out-of-specification fuel from the nuclear fuel line and any nuclear material recovery that might be anticipated as part of the nuclear fuel line including nuclear material storage and waste disposition. This includes waste following shutdown and decommissioning of nuclear fuel production and processing equipment.
3. Reactor Applicants are responsible for preparing a plan which will support used/spent nuclear fuel storage and disposition following shutdown and decommissioning of the reactor.
4. Reprocessing and recycling facility Applicants are responsible for entering into agreements with DOE or other government agencies to obtain the appropriate type and quantity of used/spent nuclear fuel required for demonstration. Applicants must explain how its

reprocessing and recycling plans apply with applicable local, state, and federal requirements, including the “Idaho Settlement Agreement,” if applicable. These agreements should address ownership of product streams and identify a plan for storage of nuclear material product(s) and associated wastes. If known, the Applicant should identify the spent nuclear fuel they plan to reprocess.

5. Reactor applicants are responsible for preparing a plan which will support the acquisition of nuclear fuel utilizing nuclear feedstock provided through DOE (i.e. High Assay Low Enriched Uranium Allocation Program, Surplus Plutonium Utilization Program) or commercial vendors.

IV. MILESTONES

The Applicant is responsible for developing specific milestones to track progress under the project. Milestones listed are DOE authorization specific. Milestones must include, at minimum:

1. Nuclear Safety Design Agreement (NSDA);
2. Preliminary design documentation;
3. Preliminary Document Safety Analysis (PDSA) (~50% design completion) for the nuclear facility(s);
4. Final design documentation;
5. Documented Safety Analysis (DSA)/Technical Safety Requirements (TSR);
6. Acceptance testing, commissioning, readiness, fueling & startup/joint test group/criticality (if applicable); nuclear material introduction (if applicable);
7. Operation; and
8. Decommissioning or transitioning under NRC license or other commercial opportunities.

V. DELIVERABLES & REPORTS

Deliverables: Project deliverables may include (ref DOE-STD-1271):

1. Nuclear Safety Design Agreement (NSDA);
2. Preliminary design documents;
3. Preliminary Documented Safety Analysis (PDSA);
4. Final design documents;
5. Document Safety Analysis (DSA);
6. Disposal or disposition of all waste generated, and management of used/spent nuclear fuel; and
7. Additional data.

Reporting: In addition to the deliverables above, Applicants will be required to submit information regarding, but not limited to:

1. Performance metric reporting which drives continuous improvement (e.g., operational/schedule, Environment, Safety and Health, Nuclear Safety, Safety System Health);

2. Facility test plan progress and status;
3. Milestone progress reporting; and
4. Project schedule.

VI. DOE REVIEW & AUTHORIZATION

DOE review and authorization of the Applicant's nuclear safety envelope and supporting documentation will establish safety requirements for a DOE authorized facility. DOE's collaborative review will follow DOE's revised regulations, guidance, and procedures and practices to expedite significantly the review, approval, and deployment of advanced reactors, nuclear fuel cycle facilities, and other nuclear facilities under the Department's jurisdiction.

VII. DECOMMISSIONING

Decommissioning encompasses all activities to remove installed facilities and return the site to its original status (e.g., as a greenfield, brownfield, etc.). Where applicable, this includes, but is not limited to, reactor core defueling, used fuel removal, final disposition of spent nuclear fuel, shipment of usable product, shipment of special nuclear material to a qualified receiver, and disposal, where possible of generated wastes. The Applicant must also ensure that qualified staff and adequate financial resources are available, that safety management programs remain in place as applicable (i.e., radiation protection, waste management, emergency preparedness, etc.), and that appropriate records are maintained.

VIII. DATA

1. Applicant Data

Applicants should limit business sensitive data (e.g., commercial or financial information that is privileged or confidential), trade secrets, proprietary, or otherwise confidential information in their application unless such information is necessary to convey an understanding of the proposed project or to comply with a requirement in the RFA. For NRIC and DOE to protect proprietary or otherwise sensitive information, the Applicant must also treat the information as confidential and properly mark it as described below.

Notice of Restriction on Disclosure and Use of Data:

Pages [list applicable pages] of this document may contain trade secrets, confidential, proprietary, or privileged information that is exempt from public disclosure. Such information shall be used or disclosed only for evaluation purposes. The Government may use or disclose any information that is not appropriately marked or otherwise restricted, regardless of source.
[End of Notice]

The header and footer of every page of the application that contains confidential, proprietary, or privileged information must be marked as follows: "Contains Trade Secrets, Confidential, Proprietary, or Privileged Information Exempt from Public Disclosure." In addition, each line or paragraph containing proprietary, privileged, or trade secret information must be clearly marked with double brackets or highlighting.

Failure to comply with these marking requirements may result in the disclosure of the unmarked information under the Freedom of Information Act (FOIA) or otherwise. These protections do not apply to information that the Applicant has released publicly or is in the public domain. For additional information on DOE's FOIA regulations, see 10 C.F.R. Part 1004.

2. Project Data

Applicants may need to grant DOE a perpetual, non-assignable license for specific uses of novel data, technical, financial or otherwise, generated during the course of the project. DOE may utilize such data to support its efforts for future authorization and safety case justifications. However, if any requested data are technical, commercial, or financial data first produced in the performance of the agreement which, if it had been obtained from and first produced by a non-federal party, would be considered a trade secret or commercial or financial information that is privileged or confidential under the meaning of 5 U.S.C. § 552(b)(4), pursuant to special OT statutory authority, such data are eligible for protection from public disclosure for up to 30 years after the data are generated or first produced ("Protected Data"). Such Protected Data must be appropriately marked as protected data when delivered to DOE as outlined in the OTA terms.

IX. PERMITTING & COMPLIANCE

The Applicant is responsible for complying with all applicable laws and regulations, and for acquiring and complying with all necessary permits. Applicable laws and regulations include, among others:

1. National Environmental Policy Act (NEPA)

The decision-making regarding authorization to proceed with a particular project under this RFA will be subject to NEPA, which requires consideration of non-speculative environmental impacts of proposed actions. In accordance with EO 14301, the Government is streamlining the application of NEPA to the extent possible. While NEPA compliance is a federal agency responsibility and the ultimate decisions remain with the Federal agency, all participants selected will, if necessary, be required to assist in the timely and effective completion of the NEPA process. If it is determined that certain records must be prepared to complete the NEPA review process (e.g., biological evaluations, categorical exclusion determinations, or environmental assessments), the participant may be required to prepare the records and bear the costs to prepare the necessary records. In addition, recipients have the option to prepare any required environmental reviews following procedures and guidance provided by the lead agency. Furthermore, recipients may elect to conduct any required environmental reviews in accordance with the procedures and guidance established by the lead agency.

X. CYBERSECURITY & IT GOVERNANCE

1. Cybersecurity Framework and Safety Integration

The Applicant must develop and execute a comprehensive Cybersecurity Plan for all onsite and offsite nuclear facilities authorized under DOE P 420.2, P 420.3, and DOE-STD-1271-2025. This plan must integrate security into every engineering, design, and operational phase, ensuring full

compliance with DOE Order 205.1D, the Office of Nuclear Energy Program Cybersecurity Plan, and FAR 52.204-21. To support the facility's safety posture, all Information Technology (IT) and Operational Technology (OT) solutions must align with NIST SP 800-53 and NIST SP 800-82 standards, with security controls coordinated and approved by the NE Safety Basis Approval Authority early in the design process.

2. Connectivity and Interconnection Security Requirements

Connectivity to Idaho National Laboratory (INL) site, other national laboratories, or DOE systems requires a mandatory **Interconnection Security Agreement (ISA)**. This agreement must define the security standards used and justify any risk-based modifications.

- **Offsite Contractors:** Must implement secure data-sharing protocols for any systems interfacing with federal or INL environments.
- **Onsite Contractors:** Must conduct all activities in strict accordance with established INL site security requirements and applicable NE directives.

3. Requirements for Land-Use Only Partners (Non-Connected)

Partners utilizing DOE land without physical or logical connection to INL networks or federal resources must maintain the integrity of their independent operating environment. The Applicant must:

- **Maintain** stringent physical and environmental security controls to protect IT and OT assets located on federal property.
- **Ensure** wireless communications or remote monitoring capabilities do not interfere with or provide unauthorized gateways to surrounding federal installations.
- **Protect** non-federal Controlled Unclassified Information (CUI) in accordance with **NIST SP 800-171** and federal safeguarding standards.

4. Governance, Documentation, and Risk Authorization

The Applicant is responsible for all IT governance, including rigorous system documentation, incident reporting, and the protection of CUI throughout the contract lifecycle.

- **Authorization:** No data exchange or system connection may occur without formal **Authorization** from the **Idaho Authorizing Official (AO)**.
- **Accountability:** The Applicant assumes full responsibility for all risks within their internal operational infrastructure and must address all cybersecurity obligations as specified in this contract.

E. Eligibility Information

All Applicants must meet the following criteria to be eligible. NRIC and DOE will use all available tools and authorities to verify eligibility and alignment with the strategic interests and foreign policies of the United States.

1. Applicants must be either:
 - a. Organized, chartered, or incorporated (or otherwise formed) under the laws of a particular state or territory of the United States or under the laws of the United States, have majority domestic ownership and control, and a physical location for business operations in the United States; or
 - b. Legally formed under the laws of a state or territory of the United States, including as a subsidiary of a foreign entity, and have a physical location for business operations in the United States; such entities will be considered foreign and must submit an explicit waiver request for foreign entity participation at the time of application.

F. Application and Submission Information

SAM.gov Registration: SAM registration is required in order to be eligible for selection. Registration must remain active throughout the term of the project.

Interested applicants must provide contact information consisting of company name, contact name, telephone number and email via email to the contacts listed below by 5:00 p.m. MDT on June 19, 2026 for initial selection period. After the initial application period, this information should be provided by all interested applicants prior to submission for the continuous selection process. Respondents will receive additional instructions and a secure link via email for submitting applications.

If you have questions, or need additional information, please contact [Sam Reiss](mailto:Sam.Reiss@inl.gov) at Samuel.Reiss@inl.gov or [AnnMarie Marshall](mailto:Ann.Marshall@inl.gov) at Ann.Marshall@inl.gov. Technical questions must be submitted in writing via email to the contacts listed above. Responses to questions will be genericized and made available to all applicants through email.

Requirements before Submission:

SAM.gov Registration (this can take up to 30 days)

Effective January 1, 2020, the System for Award Management (SAM) is the central repository for common government-wide certifications and representations required of federal solicitation awardees. As registration in SAM is required for eligibility for a federal award and registration must be updated annually, Federal agencies use SAM information to comply with award requirements and avoid increased burden and costs of separate requests for such information, unless the awardee fails to meet a federal award requirement, or there is a need to make updates to their SAM registration for other purposes.

Applicants must have an active account with SAM.gov. This includes having a Unique Entity Identifier (UEI). SAM.gov registration can take several weeks. Begin that process today. To register, go to **SAM.gov**

Entity Registration and click Get Started. On the same page, the Entity Registration Checklist also has information needed to register.

Requirements for Submission:

The Technical Volume has page limitations identified in the Technical Volume section below.

Applications may be marked with a legend identifying applicable portions of the application as containing confidential and proprietary information that may not be further disclosed.

Technical Volume

The Technical Volume is limited to 50 pages plus listed attachments which do not count towards the 50-page limit. The response shall be single-spaced pages with no less than 1-inch margins and 12-point font, formatted to fit 8.5 inch-by 11-inch paper. All documents must be written in the English language. Submissions shall be in searchable Adobe Acrobat Portable Document Format (PDF). The PDF file shall include Table 1 and Table 2 from INL Guide GDE-55179, “Nuclear Energy Launch Pad Application and Evaluation Guide.” GDE-55179 Tables do not count toward the 50-page limit. Answers to the sections in Table 2 should include all assumptions made and basis to support these assumptions. The Technical Volume shall include the following sections as a one-file submittal with each of the following sections clearly labeled:

Technical Volume Information:

Section		Description of Section Content
No.	Title	
Cover Page	Cover Page	Cover page must include: Applicant and any team members, if applicable. Clearly indicate which applicant member is the lead and will execute the determined agreements on behalf of the applicant and be responsible for all agreement performance obligations. Business size. Technical points of contact to include: salutation, last name, first name, street address, city, state, zip code, telephone, electronic mail. Administrative points of contact to include: salutation, last name, first name, street address, city, state, zip code, telephone, electronic mail. Date of submission. Applicant(s) Unique Entity Identifier (UEI) Number(s). Table of Contents.
Exe Sum	Executive Summary	The Executive Overview section shall include a high-level schedule and event-based plan for the program, including deliverables and key milestones needed to accomplish the effort requested by the Government. A summary of innovative claims and relevant

Section		Description of Section Content
No.	Title	
		experience of the key personnel and applicant team members should also be included.
1	Technology Maturity	For all areas: Include a description of your plans and approach to the development and implementation of adequate material control and accounting/ safeguards and security requirements, especially for new advanced fuel forms, including processes that utilize plutonium. ** For Qualified Test Reactors: Explanation of the reactor technology, including the key technology features of the reactor (MWe/MWt, primary coolant, secondary coolant, etc.), design progression/maturity, the Technology Readiness Level (TRL) [ref. DOE G 413.3-4A] of critical technology elements, selected manufacturer of major components including but not limited to primary vessel, fuel, reactivity control, plant logic system, power conversion system, heat exchangers, etc., and the ongoing or planned activities to complete any technology maturation actions prior to testing. Summarize probabilistic risk assessments and any technical risks and mitigation plans. For Conversion/Deconversion, Enrichment, or Fuel Fabrication Facilities: Explanation of the fuel fabrication including fuel qualification plan, conversion/deconversion or enrichment technology, including the key technology features, expected throughput and total capacity, design progression/maturity, the Technology Readiness Level (TRL) [ref. DOE G 413.3-4A] of critical technology elements, selected manufacturer of major components and ongoing or planned activities to complete any technology maturation actions prior to testing. Summarize facility hazards, technical risks, and mitigation plans. Identify the applicable standards for product acceptance. For Reprocessing & Recycling Facilities: Explanation of the recycling technology, including key technology features and expected input and product streams, design progression/maturity, the Technology Readiness Level (TRL) [ref. DOE G 413.3-4A] of critical technology elements, selected manufacturer of major components and ongoing or planned activities to complete any technology maturation actions prior to testing. Summarize probabilistic risk assessments and any technical risks and mitigation plans.
2	Nuclear Material Feedstock/Fuel	** For Qualified Test Reactors: Explanation of the plans to qualify, procure, and deliver fuel, and (following irradiation) plans for storage and disposition of used fuel. This should include the plan for fueling

Section		Description of Section Content
No.	Title	
		and defueling the reactor. The use of an agreement for used/spent fuel disposition which will outline the management of materials between DOE and applicant will be required. <u>For Conversion/Deconversion or Fuel Fabrication:</u> Explanation of the plans to procure and process feed materials and, store, and deliver converted/deconverted products and fabricated fuel material to authorized recipients and pathways for storage and disposition of waste material(s). <u>For Enrichment Facilities:</u> Explanation of the plans to procure, process, and qualify the feed materials, and, store, allocate, and deliver enriched material to authorized recipients and pathways for storage and disposition of waste material(s) <u>For Reprocessing & Recycling Facilities:</u> Explanation of the plans to procure and transport used/spent nuclear fuel to the facility, and pathways for storage and utilization or disposition of product and waste material(s).
3	Waste Management and Decommissioning	<u>Explanation of plans for waste management and decommissioning, including waste streams, defined pathways for handling and disposal, and a credible decommissioning strategy aligned with regulatory expectations.</u>
4	Facility Test Objectives	** For Qualified Test Reactors: Explanation of the reactor tests that are planned at reactor site, including the test campaigns, test objectives, test durations, needed test equipment or capabilities, operator training plans and operating plans and experimental equipment which will utilize reactor offtake (i.e. waste heat and/or electricity). Decommissioning and management of used fuel and waste management. <u>For Conversion/Deconversion, Enrichment, or Fuel Fabrication Facilities:</u> Explanation of the plan to establish the enrichment cascade, material conversion/deconversion lines, or fuel fabrication process line and the types of data that will be obtained through testing. Identify testing process (e.g., online enrichment monitoring) and applicable standards. Any need for test irradiations to qualify processes or fabricated fuel product should be discussed. Plans for both facility types should include details of expected interfaces, installation, shielding requirements, workforce development, operations, waste management, and decommissioning. <u>For Reprocessing & Recycling Facilities:</u> Explanation of the plan to establish the recycling line, including how each unit operation will be validated and what types of test campaigns and throughput will be required to achieve test objectives. Plans should include details of expected interfaces, installation, shielding requirements, workforce

Section		Description of Section Content
No.	Title	
		development, operations, waste management, and decommissioning.
5	Project Schedule	Schedule of activities from submission of the application through decommissioning of all proposed nuclear and support facility(s). The schedule should provide sufficient detail to demonstrate the applicant's lifecycle for testing and operation at the site including nuclear feedstock and/or fuel management. Applicants shall submit their schedule in the format of a Work Breakdown Structure, plus one additional level of detail to show activities that support major milestones.
6	Site Integration Plan	Explanation of how the nuclear facility interfaces with the proposed site and any needs that are beyond the current capabilities of the proposed site (e.g., site modifications, specialized equipment or tools, new facilities, storage, transportation, etc.). Launch Pad INL Applicants must include projected construction power needs, projected operational power loads, water consumption needs, and estimated land requirements specific to the facility(s) spatial requirements.
7	DOE Authorization Plan	Explanation of the basis for applying DOE authorization to the project given the project attributes, as well as an explanation of the regulatory compliance plan and activities for obtaining DOE authorization for startup, including, but not limited to, regulatory requirements for nuclear safety, quality, and environmental regulatory compliance.
8	Funding	Explanation of the funding amounts and sources that have been secured to support activities from application submission through decommissioning, including any future funding that the applicant will need to secure. Applicant cost estimates for the same time period should also be provided. Any material change to the status of this information must be reported to DOE and NRIC.
9	Internal Experience & Capabilities	Explanation of the applicant's experience, as well as the credentials of key personnel and management infrastructure (e.g., leadership, engineering design, quality assurance, procurement, project controls, etc.). In addition, describe any partnerships that will be leveraged, the nature of those partnerships. The Management Infrastructure will discuss in detail how the applicant intends to manage the overall scope of the project. The section must detail: Management of Applicant Team: Applicant Team's capability to manage the team of performers participating on the project. All entities participating as team members must be discussed in detail, along with the Applicant Team's strategy for

Section		Description of Section Content
No.	Title	
		managing and integrating the efforts of these various performers. Include short descriptions of past projects and information on whether the project was completed on time or within budget.
10	External Needs & Capabilities	Explanation of areas that require outside expertise, including from INL, other Laboratories, educational institutions, other consultants or expertise. Please identify specific areas of support and your planned source to provide or identify if support is requested to provide support. Nuclear Lifecycle Innovation Campus (NLIC): The NLIC locations may support activities across the full nuclear fuel lifecycle, including fuel fabrication, enrichment, reprocessing used nuclear fuel, and disposition of waste. Indicate if you would like to be connected with NLIC for potential opportunities. US Nuclear NEXUS: Nexus serves as a “one-stop shop” for U.S. companies seeking information or assistance from the National Nuclear Security Administration on export controls, international safeguards, security, and proliferation resistance to support US companies pursuing international deployment. Indicate if you would like to be connected with the Nuclear NEXUS.
11	Commercialization and Regulatory Plan	Explanation of the applicant’s next steps and plans for long-term commercialization. Include plans to transition from DOE authorization to commercial licensing for ongoing operations and plans for commercial licensing, future tests and demonstrations and their planned locations.
Attachment 1	Master Document and Deliverable List (MDDL)	The MDDL is a comprehensive, hierarchical catalog of all the documents and files related to the project. The MDDL includes information such as document number, revision, title, and maturity status (e.g., completion percentage). This serves as a central reference point, providing an overview of the project's documentation and helping team members find relevant and up-to-date information. Furthermore, the MDDL helps maintain organization, control, and versioning of documents.
Attachment 2	Project Schedule	See Section 4 above. Information in the Attachment will be a more in-depth schedule presented in chart form (e.g., XER, MPP).
Attachment 3	Resumes	For each proposed key personnel, applicant shall submit resumes, not to exceed two (2) pages per person proposed. Describe in sufficient and succinct detail the technical experience of the proposed individual that meets or exceeds the minimum education and experience required to perform the work.
Attachment 4	Transparency of Foreign Connections	Applicant shall submit the completed and certified Transparency of Foreign Connections document (see Sec. G “Application Review Information”). Document can be found at: Transparency of Foreign Connections Department of Energy.

Section		Description of Section Content
No.	Title	
** The term "Qualified Test Reactor" also means a nuclear fission reactor with significant improvements compared to reactors operating on December 27, 2020, including improvements such as: (i) additional inherent safety features; (ii) lower waste yields; (iii) improved fuel and material performance; (iv) increased tolerance to loss of fuel cooling; (v) enhanced reliability or improved resilience; (vi) increased proliferation resistance; (vii) increased thermal efficiency; (viii) reduced consumption of cooling water and other environmental impacts; (ix) the ability to integrate into electric applications and nonelectric applications; (x) modular sizes to allow for deployment that corresponds with the demand for electricity or process heat; and (xi) operational flexibility to respond to changes in demand for electricity or process heat and to complement integration with intermittent renewable energy or energy storage.		

G. Application Review Information

1. Basis for Selection

The Applicant may be considered only upon preparation and submittal of a fully compliant application. The selections will be based on evaluation of applications as described in this section. By submitting an application, each Applicant acknowledges and agrees that:

- NRIC and DOE may utilize the technical expertise of subject matter experts to assist in the evaluation of applications;
- Each subject matter expert who reviews any application shall be legally and contractually bound by a non-disclosure agreement governed by 41 USC § 2102 *Prohibitions on Disclosing and Obtaining Procurement Information*; and
- Each subject matter expert who reviews any application shall not be disqualified or otherwise precluded from working on other projects or agreement at his/her employer that are similar or related to the subject matter of this RFA, providing that the subject matter expert shall fully comply with the nondisclosure obligations referenced above.
- DOE shall review applications for agreements or foreign ownership, control, or influence or other risks to research, technology, and economic security, of both the entity and covered individuals.
- For the purposes of this RFA, a covered individual means an individual who (a) contributes in a substantive, meaningful way to the development or execution of the scope of work of a project proposed for funding by DOE, and (b) is designated as a covered individual by DOE. DOE designates as covered individuals any principal investigator (PI); project director (PD); coprincipal investigator (Co-PI); co-project director (Co-PD); project manager; and any individual regardless of title that is functionally performing as a PI, PD, Co-PI, Co-PD, or project manager. Status as a consultant, graduate (master's or PhD) student, or postdoctoral associate does not

automatically disqualify a person from being designated as a “covered individual” if they meet the definition in (a) above.

- The Applicant is responsible for assessing the applicability of (a) above, against each person listed on the Application. Further, the Applicant is responsible for identifying any such individual to DOE for designation as a covered individual, if not already designated by DOE as described above.
- If the work requires a security clearance, employees must have a Public Trust (suitability) clearance and must maintain that level of clearance while performing the work.
- The Applicant’s submission of a current and pending support disclosure and/or biosketch/resume for a particular person serves as an acknowledgement that DOE designates that person as a covered individual.
- DOE may further designate covered individuals during negotiations with successful Applicants or during the period of performance of the program.
- DOE in its sole discretion may utilize this information to reject applicants based on discovered security risks.

2. Application Evaluation

A review committee comprised of DOE and NRIC/INL representatives will be assembled to review the applications and make recommendations. The review committee will evaluate applications to determine based on merit which applications meet the qualifications and are selectable to participate in this program. DOE will make the ultimate decision on which applications are selected to participate in Launch Pad. DOE reserves the right to select all, some, or none of the applications received in response to this RFA. There is no maximum number of qualified applications, but DOE will prioritize projects based on the quality of the proposed project and deployment timelines. Applications received by the initial application deadline will also be prioritized.

The Review Committee shall review applications for foreign ownership, control, or influence. DOE in its sole discretion may utilize this information to reject applicants based on discovered security risks.

Merit-based selection procedures include evaluating applications based on how well the proposed activities address the following technical categories listed below:

Application Technical Adequacy Characteristics

Category	Characteristics of a Technically Complete Application
Technology Maturity Does the current technical maturity of the system present risks to the Applicant’s schedule?	The Applicant’s technology readiness plan is detailed and comprehensive. There are no significant technology maturation issues that would threaten the Applicant’s readiness. The Applicant demonstrates quality and completeness in their probabilistic risk assessment, ensuring it accurately reflects the plant’s design, operation, and potential risks. Technical risks and mitigation plans are manageable.

Category	Characteristics of a Technically Complete Application
Nuclear Feedstock/Fuel Is the plan to secure nuclear feedstock/fuel at a sufficient maturity level to enable deployment?	The Applicant has an available source to supply nuclear feedstock/fuel with manageable levels of execution risk. The plan for nuclear feedstock/fuel activities (e.g., procuring, fabricating, delivering, storing, assembling, loading, ownership of used fuel, etc.) has high likelihood of success.
Waste Management and Decommissioning Does the Applicant have an adequate plan for waste management and decommissioning?	The Applicant has a credible plan for facility decommissioning and disposal or long-term management of generated wastes and used/spent nuclear fuel.
Testing & Process Line Capability Does the Applicant's proposed plan establish fundamental technological viability?	The Applicant's test plans are sufficiently developed and there is no reason to expect detailed procedures will threaten readiness for the proposed project. If test is successful, the results would establish fundamental technological viability.
Project Schedule Does the Applicant's proposed schedule have a high likelihood of meeting the desired timeline?	The Applicant's schedule is detailed, logic driven, exhaustive of all workstreams, includes credible durations, and has manageable execution risks. Some attributes of a well-developed schedule might include: Work Breakdown Structure (WBS), Activities, and Milestones from application through decommissioning. Plus, at a minimum, one additional level of detail to show activities that support major milestones. Decommissioning activities are sufficiently detailed, and the decommissioning timeline is consistent with the overall timeline for subsequent testing (e.g., shut down, cool, disconnect, remove, decontaminate, etc.). Risk analysis with mitigations considered in schedule activities, durations, and reserve.
Site Plan Does the Applicant have a site for the facility or a solid plan toward securing a site? Are the applicant's technical interfaces well-aligned with the selected site capabilities or are interfaces a risk?	Site established. Letter of Intent (LOI), Memorandum of Understanding (MOU), or other documents confirming site selection is provided. If requesting a Launch Pad INL site indicate in Table 2 and provide reason/justification for the request and how being sited at INL aligns with your company's strategy. The applicant's understanding of the site's interface requirements is mature. There are no obvious challenges associated with delivering within those requirements.
DOE Authorization Plan Does the Applicant demonstrate an understanding of the authorization	The authorization plan is well described with a clear understanding of the regulator's requirements and processes.

Category	Characteristics of a Technically Complete Application
process, have a robust plan to navigate it, and are there any obvious, significant authorization threats?	The safety-related features of the technology are unlikely to present new hazards and phenomena that the regulator is unfamiliar with evaluating.
Funding Is the Applicant's funding sufficient to support their readiness for design, manufacturing, construction, operations, and decommissioning?	The Applicant's funding sources are secure and sufficient to execute the project through the end of testing and decommissioning. Funding is sufficient to cover the Applicant's contingency for realized risk.
Experience & Capabilities	The applicant possess the suite of needed capabilities, experience, and network to be successful on the needed timeline.
Long-term Regulatory and Commercialization Plans Does the applicant have a long-term strategy for their technology beyond DOE Authorization. Do they have a robust plan to navigate it, and are there any obvious, significant commercialization or regulatory concerns?	The regulatory and commercialization plans are well described with a clear understanding of the regulator's requirements and processes. The long-term strategy displays a comprehensive readiness and aligns with DOE's mission and strategy.
Master Document and Deliverable List (MDDL) Does the Applicant understand what documents and deliverables need to be developed to support the reactor experiment project?	The MDDL identifies most of the documents that will ultimately be needed to support the reactor operations and experiment execution. The MDDL is organized. The MDDL indicates specific information such as document number, revision, title, and maturity level.

H. Selection Information

The Selection Committee anticipates announcing initial selections of Applicants in accordance with the deadlines on the cover page. NRIC may announce further selections of participants on a rolling basis after that date. If you are selected to participate in this program, you will receive a notification from DOE and/or NRIC to finalize the applicable agreement(s).